Middle East Technical University
Department of Electrical and Electronics Engineering

EE201 – Circuit Theory I

Lecture 7: Circuit Graphs, the Incidence Matrix and Tellegen's Theorem

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Colour code used in these notes and in the slides			
current i	voltage v, e	power	nodes

1. Interconnection equations

A circuit with b branches has $2b$ unknowns: a voltage v_k and a current i_k for each branch k. The $2b$ equations that fix them come in two groups.

Group	How many	What they describe
element equations	b	*what* each element is: $v = Ri$, a source value, a dependent source, ...
interconnection equations	b	*how* the elements are connected: $n-1$ KCL and $b-n+1$ KVL equations

The interconnection equations do not depend on the elements at all. Replace a resistor by a capacitor or by a source, keep the same wiring, and KCL and KVL do not change. So, to write KCL and KVL, we may forget the elements and keep only the wiring. What is left is the **graph** of the circuit.

This lecture builds the graph of a circuit and writes *all* of its KCL and KVL equations with one matrix. Tellegen's theorem then follows in one line. The same bookkeeping is how a circuit simulator stores a circuit: a list of elements, each with the nodes it joins.

2. The graph of a circuit

Circuit graph

The **graph** of a circuit is the circuit with every two-terminal element replaced by a line segment, called a **branch**. The points where branches meet are the **nodes**. Each branch gets a number and an arrow.

The graph records only *which nodes each branch joins*, and in which direction it is taken. Lengths, angles and the shape of the drawing carry no information.

2.1. Procedure

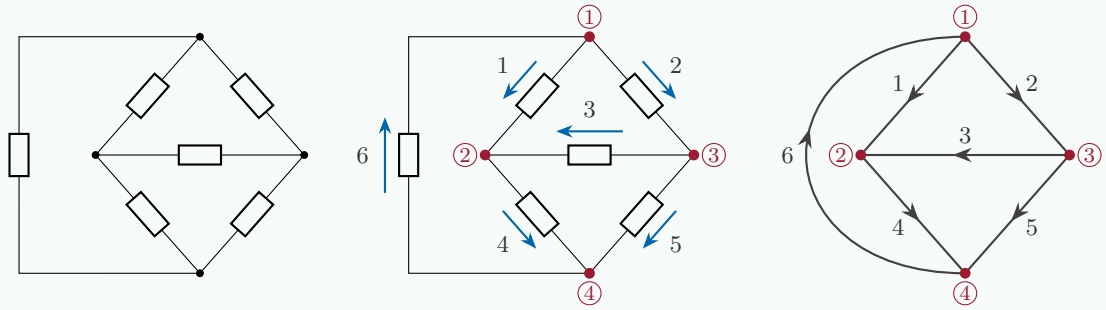
1. Label each branch: $1, 2, \ldots, b$.
2. Assign a direction to each branch. The choice is **arbitrary**.
3. Label each node: ①, ②, ..., ⓝ.
4. Draw the graph: a dot for each node and an arrow for each branch.

The arrow on branch k is the **reference direction** of its current i_k. As before we use associated reference directions, so the branch voltage v_k is measured with $+$ at the tail of the arrow and $-$ at its head. Choosing an arrow does not claim that the current really flows that way: if it flows the other way, i_k comes out negative.

2.2. Example

Example 7.1 — the graph of a bridge circuit

The circuit $\mathcal{N}$ below has six two-terminal elements, drawn as boxes because their type does not matter here. Find its graph.



(a) circuit $\mathcal{N}$ (b) steps 1–3 (c) step 4: the graph

Steps 1–3 label the six branches, give each one an arrow (blue: the current reference), and label the four nodes. Step 4 keeps only the dots and the arrows. The graph of $\mathcal{N}$ has

$$\text{nodes } \{①, ②, ③, ④\}, \quad n = 4; \qquad \text{branches } \{1, 2, 3, 4, 5, 6\}, \quad b = 6.$$

Branch 6 is drawn curved only to keep it clear of the others; a straight arrow from ④ to ① would be the same graph.

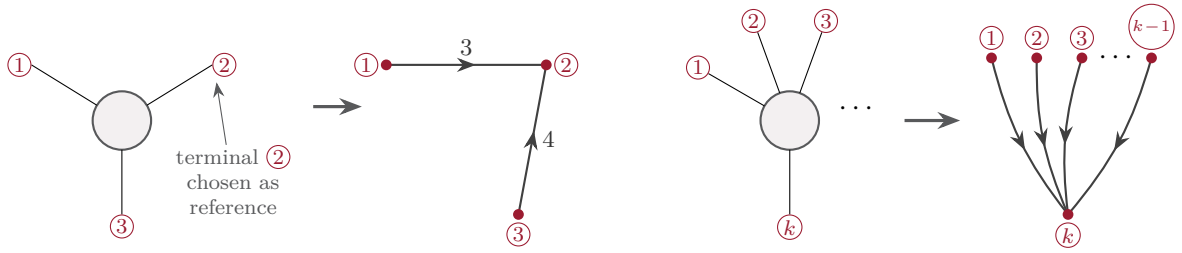
3. Elements with more than two terminals

A two-terminal element becomes one branch. Some elements have more terminals: a transistor has three, an integrated circuit can have many. Such an element cannot be one line segment.

A k-terminal element in the graph

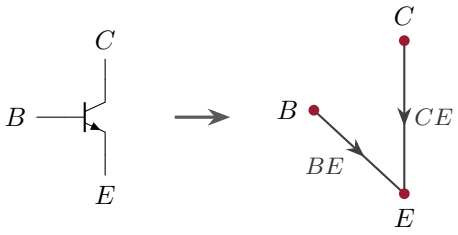
For an element with $k \geq 3$ terminals, choose one terminal as its **reference** (the choice is arbitrary). Then draw $k-1$ branches, one from each of the other terminals to the reference terminal.

Why $k-1$ branches, not k? A closed surface around the element is crossed by k terminal currents, and KCL on that surface says they sum to zero. Only $k-1$ of them are free; the last one follows. The voltages work the same way: the $k-1$ voltages measured from the reference terminal fix every other voltage between terminals, by KVL. So $k-1$ branches, each with a current and a voltage, describe the element completely.



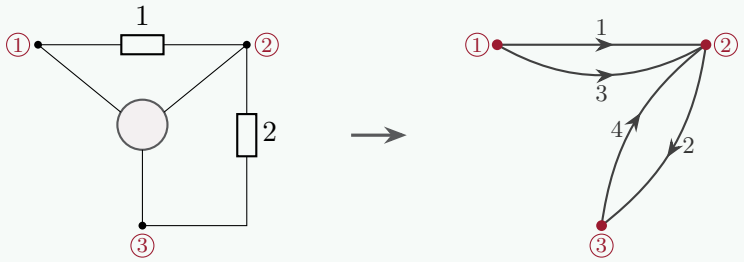
a 3-terminal element $k-1=2$ branches a k-terminal element $k-1$ branches

A transistor is the most common case. With its emitter E as the reference terminal it becomes two branches: base to emitter, and collector to emitter.



Example 7.2 — a circuit with a 3-terminal element

In the circuit below, element 1 joins nodes ① and ②, element 2 joins ② and ③, and a 3-terminal element touches all three nodes. Find the graph.



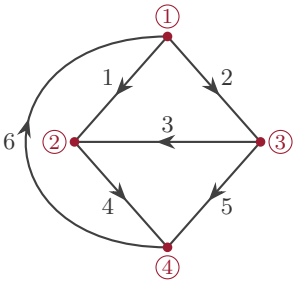
the circuit its graph: $n = 3$, $b = 4$

Elements 1 and 2 give one branch each. For the 3-terminal element choose the terminal at ② as reference; it gives branch 3 (from ① to ②) and branch 4 (from ③ to ②). Branches 1 and 3 join the same two nodes; that is allowed, and so is the pair 2 and 4.

A dependent source needs nothing new: it is already drawn as two branches, the branch that senses the controlling variable and the branch of the source.

4. KCL on a circuit graph

Every branch current flows along its own arrow. At each node, count a current **leaving** the node with + and a current **entering** it with −; KCL says the sum is zero. For the graph of $\mathcal{N}$ in Example 7.1:



$$
\begin{aligned}
① &: \quad i_1 + i_2 - i_6 = 0 \\
② &: \quad -i_1 - i_3 + i_4 = 0 \\
③ &: \quad -i_2 + i_3 + i_5 = 0 \\
④ &: \quad -i_4 - i_5 + i_6 = 0
\end{aligned}
$$

Nothing about the elements was used: the four equations come from the graph alone. Two circuits with the same graph have the same KCL equations, whatever their elements are.

5. The incidence matrix

The four equations above contain only the coefficients $+1$, -1 and 0. Collect them in a matrix, and collect the currents in a vector:

$$\begin{array}{c} \\ ① \\ ② \\ ③ \\ ④ \end{array}\!\!\begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 1 \end{bmatrix} \end{array} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \\ i_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \qquad \Longleftrightarrow \qquad A_a\,\mathbf{i} = \mathbf{0}.$$

Rows are labelled by nodes, columns by branches. A matrix that stores the graph like this is a **circuit matrix**. There are several; the first and most basic is this one.

Incidence matrix

For a graph with n nodes and b branches, the **incidence matrix** A_a is the $n \times b$ matrix with entries

$$a_{ij} = \begin{cases} +1 & \text{if branch } j \text{ leaves node } i, \\ -1 & \text{if branch } j \text{ enters node } i, \\ 0 & \text{otherwise.} \end{cases}$$

With the **branch current vector** $\mathbf{i} = [\, i_1 \; i_2 \; \cdots \; i_b \,]^T$, KCL at every node of the circuit is the single equation $A_a\,\mathbf{i} = \mathbf{0}$.

The subscript a stands for *all*: every node has a row. The matrix can be read in two directions.

- **A row** is a node. Its nonzero entries are the branches that touch that node, and the row times $\mathbf{i}$ is the KCL equation at that node.
- **A column** is a branch. It has exactly one $+1$ (the node the branch leaves) and exactly one -1 (the node it enters). Column 3 above has $+1$ in row ③ and -1 in row ②: branch 3 goes from ③ to ②.

The columns hold all of the graph: from A_a alone one can redraw the graph, arrows included. Reversing the arrow of a branch flips the sign of its column, and nothing else.

6. Review: linear dependence

Linear dependence

Let $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$ be vectors of the same size. If there are real numbers $\alpha_1, \alpha_2, \ldots, \alpha_k$, **not all zero**, with

$$\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_k\mathbf{v}_k = \mathbf{0},$$

the vectors are **linearly dependent**. Otherwise — if the only such numbers are $\alpha_1 = \cdots = \alpha_k = 0$ — they are **linearly independent**.

If the vectors are dependent, at least one of them is a combination of the others: pick an $\alpha_m \neq 0$ and solve for $\mathbf{v}_m$.

Example 7.3 — dependent or not?

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \; \begin{bmatrix} 0 \\ -1 \end{bmatrix} \text{ are independent;} \qquad \mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \; \mathbf{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \; \mathbf{v}_3 = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix} \text{ are dependent.}$$

In the first pair, $\alpha_1[1\ 0]^T + \alpha_2[0\ -1]^T = [\alpha_1 \;\; -\alpha_2]^T$, which is zero only if $\alpha_1 = \alpha_2 = 0$. In the second set, $\mathbf{v}_1 - 2\mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$ (check one entry: $1 - 10 + 9 = 0$ in the middle row), so $\mathbf{v}_3 = 2\mathbf{v}_2 - \mathbf{v}_1$. Any nonzero multiple works as well, for example $3\mathbf{v}_1 - 6\mathbf{v}_2 + 3\mathbf{v}_3 = \mathbf{0}$.

For a set of equations the meaning is direct. If the rows of a matrix M are linearly dependent, one equation of $M\mathbf{x} = \mathbf{0}$ is a combination of the others: it follows from them and adds no information. The largest number of independent rows is the **rank** of M. When all rows are independent, M has **full row rank**.

7. The reduced incidence matrix

Every column of A_a has one $+1$ and one -1. So the sum of all rows of A_a is the zero row:

$$\text{row } ① + \text{row } ② + \cdots + \text{row } ⓝ = [\,0\ 0\ \cdots\ 0\,].$$

This is the definition of linear dependence with every $\alpha = 1$. The rows of A_a are **always** linearly dependent, and one of the n KCL equations is redundant — the fact met earlier, now in matrix form.

Reduced incidence matrix

Choose one node as the **reference node** (or **ground**). Delete its row from A_a. The $(n-1)\times b$ matrix that remains is the **reduced incidence matrix**, written A.

The choice of the reference node is arbitrary. It is the same node that will serve as the common reference for node voltages.

Example 7.4 — the reduced incidence matrix of $\mathcal{N}$

Take ④ as the reference node and delete its row:

$$A_a = \left[\begin{array}{cccccc} 1 & 1 & 0 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & -1 & -1 & 1 \end{array}\right] \xrightarrow{\text{delete row } ④} A = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \end{bmatrix}.$$

A is 3×6. Its rows are independent: column 4 has a nonzero entry only in row ②, column 5 only in row ③, and column 6 only in row ①. A combination of the rows that gives zero in these three columns must have all three α's zero.

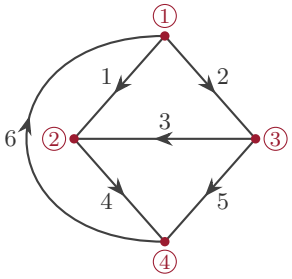
Deleting the row loses nothing.

- **No equation is lost.** If $A\,\mathbf{i} = \mathbf{0}$, then $A_a\,\mathbf{i} = \mathbf{0}$ as well: the deleted row is minus the sum of the rows that were kept, so its equation is minus the sum of theirs.
- **No part of the graph is lost.** Each column of A_a sums to zero, so the deleted row can be rebuilt from A. A column of A with a single nonzero entry is a branch that touches the reference node.

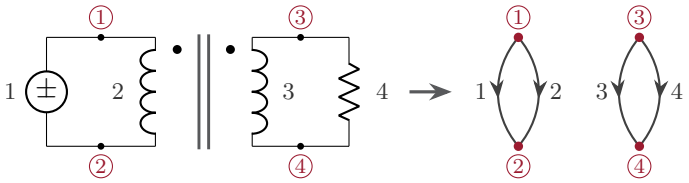
8. Connected graphs

Connected graph

A graph is **connected** if any two nodes are joined by a path of branches, ignoring the arrow directions. Otherwise it falls into separate pieces.



graph of $\mathcal{N}$: connected



a transformer circuit and its graph: two pieces

The graph of $\mathcal{N}$ is connected. The transformer circuit is not: the windings are coupled only through the core, so no branch joins the two sides. (An ideal transformer is a two-port: one branch per winding.) Other circuits split the same way — for example, two parts that interact only through a dependent source that senses in one part and acts in the other, the model of an isolation amplifier or an optocoupler.

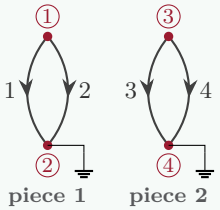
Connected graph: the rows of A are independent

If the graph is connected, the $n-1$ rows of A are linearly independent: $\operatorname{rank} A = n-1$.

8.1. A graph in separate pieces

Treat each piece as a graph of its own. Each piece is connected, so it gets its own incidence matrix, its own reference node and its own reduced incidence matrix. Then stack the reduced matrices into one.

Example 7.5 — the transformer circuit



incidence matrix **reference** **reduced**

piece 1 $\begin{matrix}\\ ① \\ ②\end{matrix}\begin{bmatrix}1 & 2\\ 1 & 1\\ \cancel{1} & \cancel{1}\end{bmatrix}$ delete row ② $A_1 = ①\begin{bmatrix}1 & 2\\ 1 & 1\end{bmatrix}$

piece 2 $\begin{matrix}\\ ③ \\ ④\end{matrix}\begin{bmatrix}3 & 4\\ 1 & 1\\ \cancel{1} & \cancel{1}\end{bmatrix}$ delete row ④ $A_2 = ③\begin{bmatrix}3 & 4\\ 1 & 1\end{bmatrix}$

Stack: $A = \begin{bmatrix}A_1 & 0\\ 0 & A_2\end{bmatrix} = \begin{matrix}\\ ① \\ ③\end{matrix}\begin{bmatrix}1 & 2 & 3 & 4\\ 1 & 1 & 0 & 0\\ 0 & 0 & 1 & 1\end{bmatrix}$ $\quad i_1 + i_2 = 0, \quad i_3 + i_4 = 0.$

Two independent KCL equations: $n - p = 4 - 2 = 2$.

With p separate pieces there are p reference nodes, one per piece, and A has $n-p$ independent rows. In EE201 graphs are connected unless stated otherwise.

9. KCL in matrix form

KCL in matrix form

For a circuit whose graph is connected, with n nodes and b branches,

$\boxed{A\,\mathbf{i} = \mathbf{0}}$ $\quad A \in \mathbb{R}^{(n-1)\times b}$: reduced incidence matrix, $\quad \mathbf{i} \in \mathbb{R}^{b}$: branch current vector.

These are $n-1$ **independent** KCL equations, and every other KCL equation of the circuit follows from them. For a graph in p separate pieces, build A piece by piece; it has $n-p$ rows.

Example 7.6 — KCL for the circuit with a 3-terminal element

Write A_a, A and the KCL equations for the graph of Example 7.2, with ③ as the reference node.

The arrows are: branch 1 from ① to ②, branch 2 from ② to ③, branch 3 from ① to ②, branch 4 from ③ to ②. Column by column:

$$A_a = \begin{array}{c} \\ ① \\ ② \\ ③ \end{array}\!\!\begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 1 & -1 & -1 \\ 0 & -1 & 0 & 1 \end{bmatrix} \end{array} \qquad A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 1 & -1 & -1 \end{bmatrix},$$

$$A\,\mathbf{i} = \mathbf{0}: \qquad i_1 + i_3 = 0, \qquad -i_1 + i_2 - i_3 - i_4 = 0.$$

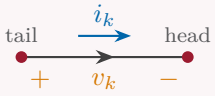
Columns 1 and 3 are equal because branches 1 and 3 join the same nodes in the same direction. The first equation says the current leaving ① through element 1 returns through the element's terminal at ①. The reference terminal of the element (at ②) and the reference node of the graph (③) are two separate choices.

10. Branch voltages and node voltages

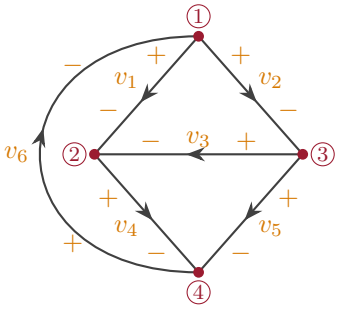
The incidence matrix also gives KVL. For that we need two kinds of voltage.

Branch voltages and node voltages

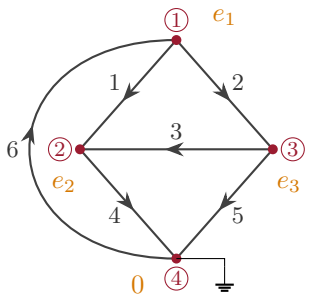
Branch voltage v_k: the voltage across branch k, + at the tail of the arrow. So i_k enters at + (associated reference).



Node voltage e_k: the voltage of node k with respect to the reference node. The reference node has $e = 0$.



branch voltages $\mathbf{v} = [\,v_1 \;\cdots\; v_6\,]^T$



node voltages $\mathbf{e} = [\,e_1\;\; e_2\;\; e_3\,]^T$, reference ④

$b = 6$ branch voltages, but only $n - 1 = 3$ node voltages. We write v_{pq} for the voltage of node p (+) with respect to node q (−). So $e_1 = v_{14}$, $e_2 = v_{24}$, $e_3 = v_{34}$, and $v_{qp} = -v_{pq}$.

11. KVL on a circuit graph

By KVL, every branch voltage can be written in terms of node voltages. Branch 1 of $\mathcal{N}$ goes from ① to ②, so $v_1 = v_{12}$. Go from ① to ② through the reference node ④:

$$v_1 = v_{12} = v_{14} + v_{42} = v_{14} - v_{24} = e_1 - e_2.$$

In general, a branch from node p to node q has $v_k = e_p - e_q$: **tail minus head**. If one end is the reference node, its node voltage is 0. For all six branches of $\mathcal{N}$:

$$\left.\begin{aligned} v_1 &= e_1 - e_2 \\ v_2 &= e_1 - e_3 \\ v_3 &= e_3 - e_2 \\ v_4 &= e_2 \\ v_5 &= e_3 \\ v_6 &= -e_1 \end{aligned}\right\} \quad \Longrightarrow \quad \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix}}_{A^T} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}.$$

The matrix is A^T, the transpose of A from Example 7.4: its row k is column k of A. For example, row 1 is $[\,1 \;\; -1 \;\; 0\,]$, and column 1 of A is $[\,1 \;\; -1 \;\; 0\,]^T$.

KVL in matrix form

$\boxed{\mathbf{v} = A^T \mathbf{e}}$ $\quad \mathbf{v} \in \mathbb{R}^b$: branch voltage vector, $\quad \mathbf{e} \in \mathbb{R}^{n-1}$: node voltage vector.

The same matrix A gives KCL ($A\mathbf{i} = \mathbf{0}$) and KVL ($\mathbf{v} = A^T\mathbf{e}$).

Remarks.

- Once the $n-1$ node voltages are known, all b branch voltages are known.
- KVL puts **no constraint** on the node voltages: any $\mathbf{e}$ gives branch voltages that satisfy KVL around every loop. Around a loop the differences cancel. For the loop of branches 1, 3, 2: $v_1 - v_3 - v_2 = (e_1 - e_2) - (e_3 - e_2) - (e_1 - e_3) = 0$.

So the node voltages are a good choice of unknowns: there are only $n-1$ of them, they are free, and KVL is then satisfied automatically. This is the idea behind node analysis.

Example 7.7 — KVL for the circuit with a 3-terminal element

For the graph of Example 7.6 (reference ③), A has two rows, so there are two node voltages e_1 and e_2:

$$\mathbf{v} = A^T\mathbf{e}: \quad \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 1 & -1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} e_1 - e_2 \\ e_2 \\ e_1 - e_2 \\ -e_2 \end{bmatrix}.$$

Branches 1 and 3 join the same two nodes, so they have the same voltage. Branch 4 goes from ③ (the reference) to ②, so $v_4 = 0 - e_2$.

12. Tellegen's theorem

KCL and KVL together give a short but powerful result.

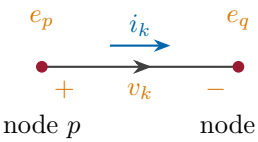
Tellegen's theorem

In any circuit, with associated reference directions,

$$\mathbf{i}^T\mathbf{v} = \sum_{k=1}^{b} v_k\, i_k = 0 \qquad \text{at every instant } t.$$

$v_k i_k$ is the power absorbed by branch k. So the theorem says the total power absorbed by all branches is zero: the power delivered by some branches is absorbed by the others. This is **conservation of power**.

Proof. Take branch k from node p to node q.



By KVL its voltage is tail minus head, $v_k = e_p - e_q$, so its power splits into two terms, one for each end:

$$v_k i_k = \underbrace{e_p i_k}_{i_k \text{ leaves } p} - \underbrace{e_q i_k}_{i_k \text{ enters } q} .$$

Add over all branches and collect the terms of each node n. Every branch that leaves n contributes $+e_n i_k$ and every branch that enters n contributes $-e_n i_k$:

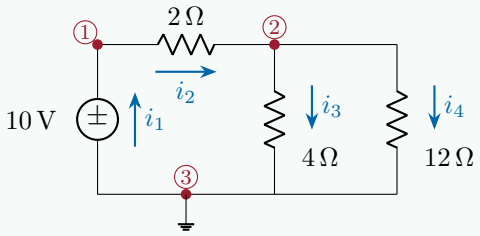
$$\sum_k v_k i_k = \sum_n e_n \Big(\underbrace{\textstyle\sum \text{currents leaving } n - \sum \text{currents entering } n}_{=0 \text{ by KCL}} \Big) = 0.$$

(The reference node has $e = 0$, so it adds nothing.) For $\mathcal{N}$ the brackets are exactly the three KCL equations of Section 4: $\sum_k v_k i_k = e_1(i_1 + i_2 - i_6) + e_2(-i_1 - i_3 + i_4) + e_3(-i_2 + i_3 + i_5)$. In matrix form the two steps are one line: $\mathbf{i}^T\mathbf{v} \overset{\text{KVL}}{=} \mathbf{i}^T A^T \mathbf{e} = (A\mathbf{i})^T \mathbf{e} \overset{\text{KCL}}{=} 0$. ■

The proof uses KCL and KVL only — never the element equations. That is why knowing that $\mathbf{i}$ and $\mathbf{v}$ come from the same circuit does not make it any shorter, and why the theorem holds even more generally (see the appendix).

Example 7.8 — checking Tellegen's theorem

Find the voltage and the current of every branch with node voltages, and check that $\sum_k v_k i_k = 0$. Each branch has the current reference shown; its voltage has + at the tail of that arrow. Reference node: ③.



1. Node voltages. The source fixes $e_1 = 10\,\text{V}$; e_2 is the only unknown.

2. KCL at ②, with each current written by Ohm's law in terms of node voltages ($-i_2 + i_3 + i_4 = 0$):

$$-\frac{e_1 - e_2}{2} + \frac{e_2}{4} + \frac{e_2}{12} = 0 \quad \xrightarrow{\times 12} \quad -6(10 - e_2) + 3e_2 + e_2 = 0 \quad \Rightarrow \quad e_2 = 6\,\text{V}.$$

3. Branch voltages, $\mathbf{v} = A^T\mathbf{e}$ (tail minus head): $v_1 = 0 - e_1 = -10\,\text{V}$, $v_2 = e_1 - e_2 = 4\,\text{V}$, $v_3 = v_4 = e_2 = 6\,\text{V}$.

4. Branch currents, Ohm's law: $i_2 = 4/2 = 2\,\text{A}$, $i_3 = 6/4 = 1.5\,\text{A}$, $i_4 = 6/12 = 0.5\,\text{A}$. KCL at ①: $-i_1 + i_2 = 0$, so $i_1 = i_2 = 2\,\text{A}$.

5. Check.

branch	element	v_k (V)	i_k (A)	$v_k i_k$ (W)
1	source	−10	2	−20
2	2 Ω	4	2	8
3	4 Ω	6	1.5	9
4	12 Ω	6	0.5	3
			sum	0

Branch 1 goes from ③ to ①, so the + of v_1 is at the bottom and $v_1 = -10\,\text{V}$: $v_1 i_1 = -20\,\text{W}$. The source delivers 20 W, and the resistors absorb $8 + 9 + 3 = 20\,\text{W}$.

Summary

- The $2b$ equations of a circuit split into b **element equations** and b **interconnection equations** (KCL, KVL). The second group depends only on the wiring, that is, on the **graph**.
- The graph: number the branches, give each an arbitrary arrow (the current reference), number the nodes, and keep only dots and arrows. A k-terminal element becomes $k - 1$ branches to a reference terminal.
- The **incidence matrix** A_a ($n \times b$): +1 where a branch leaves a node, −1 where it enters. KCL at all nodes is $A_a\mathbf{i} = \mathbf{0}$. Each column has one +1 and one −1, so the rows sum to zero and are dependent.
- The **reduced incidence matrix** A ($(n-1) \times b$): A_a without the reference node's row. For a **connected** graph its rows are independent, and $A\mathbf{i} = \mathbf{0}$ gives exactly the $n - 1$ independent KCL equations. For a graph in separate pieces: one reference node per piece, then stack.
- **Node voltages** $\mathbf{e}$ ($n - 1$ of them, measured from the reference). KVL: $\mathbf{v} = A^T\mathbf{e}$, each branch voltage is tail minus head. KVL puts no constraint on $\mathbf{e}$.
- **Tellegen's theorem**: in any circuit $\mathbf{i}^T\mathbf{v} = \sum v_k i_k = 0$ — conservation of power, from KCL and KVL alone. (A general form is in the appendix.)

Next lecture

Planar graphs and meshes: the **mesh matrix** M, KVL as $M\mathbf{v} = \mathbf{0}$, and mesh currents with $\mathbf{i} = M^T\mathbf{i}_M$ — the counterpart of $\mathbf{v} = A^T\mathbf{e}$.

Reading

- Alexander & Sadiku, *Fundamentals of Electric Circuits*, Section 2.4 (nodes, branches and loops).
- Chua, Desoer & Kuh, *Linear and Nonlinear Circuits* — circuit graphs, the incidence matrix, KCL and KVL in matrix form, Tellegen's theorem.
- Lecture notes and videos of Prof. Emre Tuna, `users.metu.edu.tr/etuna/ee201` — notes pp. 12–15. Lecture 6 video: interconnection equations 2:18, circuit graph 4:00–7:55, k-terminal elements 13:45, KCL on the graph 21:49, circuit matrices and the incidence matrix 26:04–30:40, linear dependence 35:19, reduced incidence matrix 40:45, connected graphs 46:52, summary 50:15. Lecture 7 video: branch and node voltages 0:35–6:57, KVL on the graph 8:30, remark 17:55, Tellegen's theorem 19:39, conservation of power 24:31.

Appendix: Tellegen's theorem, general form

The proof used only two facts: $\mathbf{i}$ satisfies KCL and $\mathbf{v}$ satisfies KVL, both with the same A. Nothing about the elements was used, and nothing requires $\mathbf{i}$ and $\mathbf{v}$ to come from the same circuit.

Tellegen's theorem, general form

Take two circuits with the **same graph**, with any elements. Let $\mathbf{i}$ be the branch currents of circuit 1 and $\mathbf{v}$ the branch voltages of circuit 2. Then $\mathbf{i}^T\mathbf{v} = 0$.

Example A.1 — the general form on the graph of $\mathcal{N}$

Take any branch currents that satisfy KCL on the graph of $\mathcal{N}$, and any node voltages — they do not have to come from the same circuit:

$$\mathbf{i} = [\,1 \;\; 2 \;\; 1 \;\; 2 \;\; 1 \;\; 3\,]^T\,\text{A}, \qquad \mathbf{e} = [\,5 \;\; 2 \;\; 3\,]^T\,\text{V}.$$

KCL holds: $1 + 2 - 3 = 0$, $-1 - 1 + 2 = 0$, $-2 + 1 + 1 = 0$. The branch voltages are $\mathbf{v} = A^T\mathbf{e} = [\,3 \;\; 2 \;\; 1 \;\; 2 \;\; 3 \;\; -5\,]^T\,\text{V}$. Then

$$\mathbf{i}^T\mathbf{v} = 1\cdot 3 + 2\cdot 2 + 1\cdot 1 + 2\cdot 2 + 1\cdot 3 + 3\cdot(-5) = 15 - 15 = 0.$$